

DELFT UNIVERSITY OF TECHNOLOGY

LITERATURE REVIEW

MARINE TECHNOLOGY

Understanding Prolate Jellyfish Locomotion Efficiency through Numerical Simulation of Bell Kinematics.

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Abstract

This literature review focuses on understanding and modelling the morphology, kinematics and hydrodynamics of jellyfish. The biomechanics of jellyfish show that these organisms utilise complex hydrodynamic processes that benefit propulsive efficiency. These hydrodynamic processes are suction thrust, jet expulsion, passive energy recapture and the virtual wall effect, all directly associated with the generation of starting and stopping vortices. Various approaches for numerically modelling the jellyfish and the kinematics of its bell are shown to set a base for new research progress in this field. In general, methods are implemented that make several assumptions regarding dimensionality and bell margin motion. Bell margin motions have been implemented actively and passively to search for the most biologically accurate model. The search for the most efficient kinematics for jellyfish locomotion reveals a research gap in 3D-modelling of jellyfish morphology with actively controlled bell margin motion. This provides the basis for this research on jellyfish locomotion efficiency, with a 3D shape model with actively controlled bell margin motion, through CFD simulations with WaterLily.jl.

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1 Introduction

1.1 Motivation

Despite their unique and specific anatomical structures, jellyfish exhibit some of the highest propulsive efficiencies recorded in the animal kingdom, in terms of scaled swimming speed and cost of transport (Baldwin & Battista, 2021; Hoover et al., 2017). Rhythmic bell contractions and the generation of vortex rings are characteristic of jellyfish swimming. Jellyfish use fluid dynamic interactions like suction thrust, passive energy recapture, and virtual wall effects for locomotion (Costello et al., 2021; Dabiri et al., 2005). This forward motion is achieved while having limited muscle thickness and minimal neural complexity. (Costello et al., 2008; Park et al., 2014).

Although jellyfish locomotion involves complex fluid–structure interactions that can be challenging to model, their anatomy and accessibility for observation make them ideal research subjects. Existing jellyfish research concerns the numerical modelling of hydrodynamics and the development of biomimetic propulsion systems (Costello et al., 2021; Frame et al., 2018; Xiao et al., 2013). Although advances in available CFD methods and modelling have been made, designing accurate jellyfish models remains difficult. General challenges are model construction, convergence reliability, and simulation time (Park et al., 2014). As a result, a detailed understanding of how specific kinematics and active flow control strategies contribute to their high swimming efficiency remains incomplete (Costello et al., 2021).

1.2 Knowledge Gap

A limitation across the literature is the absence of a non-simplified three-dimensional bell model, with a separately moving bell margin. All research applies simplification in some way, either through shape simplification (assuming a geometric parameterisation) or reduced dimensionality. Detailed morphological accuracy, in terms of having a flap and tentacles, is lacking, although several simplified options have been presented and discussed.

The flow control mechanism of the jellyfish bell margin has been modelled as both active and passive across different studies. As a result, several different models exist, either applying active control through prescribed kinematics or passive control through fluid-structure interaction. No accurate 3D model, including the control of this flap, is currently available. Wilson and Eldredge (2011) created a simplified model and stated that passive control is most efficient. However, the influence of bell margin kinematics on the propulsive efficiency of jellyfish has not been assessed with an accurate 3D model.

Achieving a validated 3D numerical jellyfish morphology model that can biologically accurately mimic jellyfish kinematics, including a flow control mechanism for separate motion of the bell margin, would be a significant advancement in current modelling approaches. Determining the optimal bell margin kinematics to achieve the highest propulsive efficiency would provide new insights in this field of research. As experimental hydrodynamic data is available for hydromedusae in Colin and Costello (2002), it makes sense to choose this type of jellyfish. To avoid the complexity of (semi-)passive bell margin modelling and make the use of WaterLily.jl as a solver possible, the choice is made to control bell margin motion actively.

1.3 Scope

The core objective of this research is to deepen the understanding of the kinematic and fluid dynamic phenomena underlying prolate jellyfish locomotion and how these mechanisms contribute to their high propulsion efficiency. The choice for the prolate type results from the lower Reynolds number, making it more appropriate for simulations with flow solvers. This project aims to model the 3D shape and kinematics of a prolate jellyfish in a biologically accurate way by validating this model with available experimental jellyfish data in Sahin et al. (2009). Furthermore, an active flow control mechanism will be applied to control the bell margin kinematics as a means to investigate its influence on the propulsive efficiency of the jellyfish. Computational fluid dynamics will be applied to simulate the jellyfish and generate efficiency metrics.

An existing 2D numerical jellyfish model, developed by Dr. Artur Lidtke, will serve as the foundation for research on jellyfish efficiency. This model numerically represents the flapping motion of the jellyfish bell responsible for jellyfish locomotion (Costello et al., 2021). Using motion data derived from the literature, this model will be adjusted to match known behaviours of both kinematics and fluid dynamics. Hydrodynamic simulations will be performed using WaterLily, a flow simulator developed by Weymouth and Font (2025). A reinforcement learning algorithm is studied as a possible control mechanism for the bell kinematics in these simulations. This enables the possibility of generating numerous situations in relatively little time, creating an extensive data set of results. Optimal bell kinematics can be derived in different flow simulations, offering broader insights into biologically optimised propulsion strategies.

1.4 Research Questions

A main research question was determined based on the described knowledge gap in the available literature:

“Why do prolate jellyfish have high locomotion efficiency, and how can an experimentally-validated numerical jellyfish model be developed to investigate this?”

The following objectives are defined to answer the main research question.

1. Design a 3-dimensional numerical model in WaterLily that accurately describes prolate jellyfish bell kinematics.
2. Apply an active flow control mechanism that can move the bell margin separately from the bell.
3. Determine the kinematics settings that acquire the highest jellyfish propulsive efficiency / lowest cost of transport.

2 Literature Review

2.1 Biomechanics of jellyfish locomotion

Before presenting the state of the art through a literature review, a theoretical framework is given to establish a basic understanding of jellyfish characteristics. Key anatomical, kinematic, and hydrodynamic concepts are introduced. Anatomy refers to the physical structures involved in movement. The kinematics describe how jellyfish move their physical structures during swimming cycles. The hydrodynamics explain how these movements generate thrust for jellyfish locomotion.

2.1.1 Jellyfish Anatomy

Medusae belong to the animal phylum Cnidaria, part of the Animalia kingdom. Medusae are part of the Medusozoa subphylum, which exhibits a biphasic life cycle consisting of a polyp stage and a free-swimming medusa stage. Medusae in the free-swimming stage are commonly referred to as jellyfish (Costello et al., 2021; Leclère & Röttinger, 2017). The diversity of the Medusozoa subphylum, in terms of size, shape, body design and propulsive mode, leads to another subdivision into classes, presented in Figure 1. Frequently studied classes in the field of kinematics and hydrodynamics are the Scyphozoa and Hydrozoa. *Aurelia aurita* (moon jellyfish) is an example species within this genus that has been widely investigated from the point of view of propulsion, because of its high propulsive efficiency (Villanueva et al., 2014).

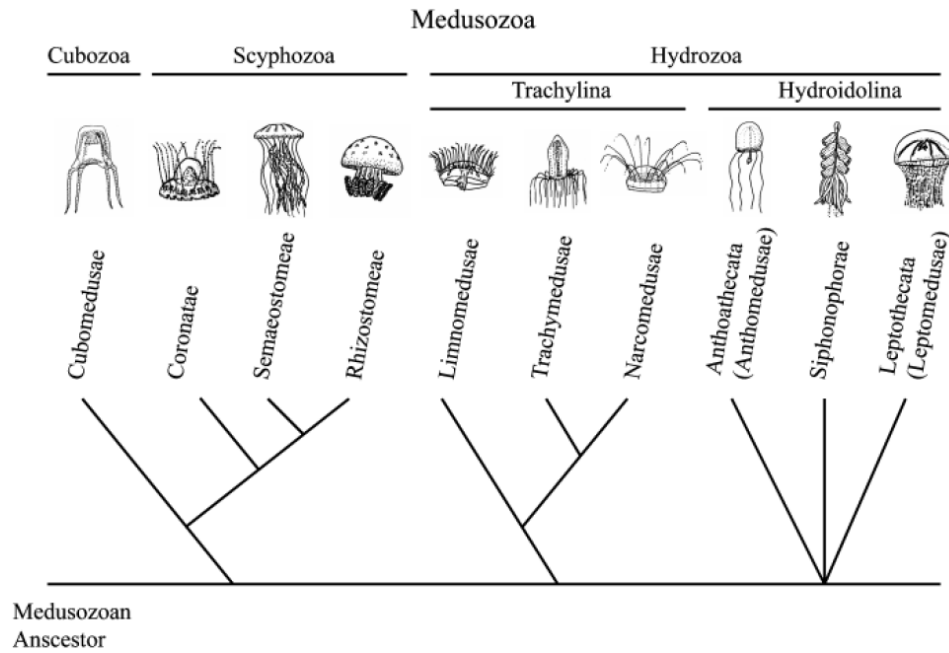


Figure 1: Medusan Diversity across the medusozoa subphylum. Figure reproduced from (Costello et al., 2008)

Medusozoa jellyfish swim through rhythmic contractions of a muscular bell (Baldwin & Battista, 2021), but are distinguished by the presence (Hydrozoa) or absence (Scyphozoa) of a velum, which affects the mode of propulsion. A velum is a muscular, flap-like structure connected to the bell margin. Jellyfish, having a velum, typically have a prolate bell shape, applying jet propulsion for locomotion. Jellyfish, which do not have a velum, are generally oblate and often use a rowing mode of propulsion (Costello et al., 2008, 2021; Daniel, 1995). These propulsion

modes are further explained in Section 2.1.2. Oblate bell shapes are characterised as a curved plate, tending to have a smaller fineness ratio, which is the ratio between bell height and bell diameter. Prolate jellyfish are generally small with a greater fineness ratio, shaped like a rocket (Dabiri et al., 2007; Gemmell, Du Clos, et al., 2021).

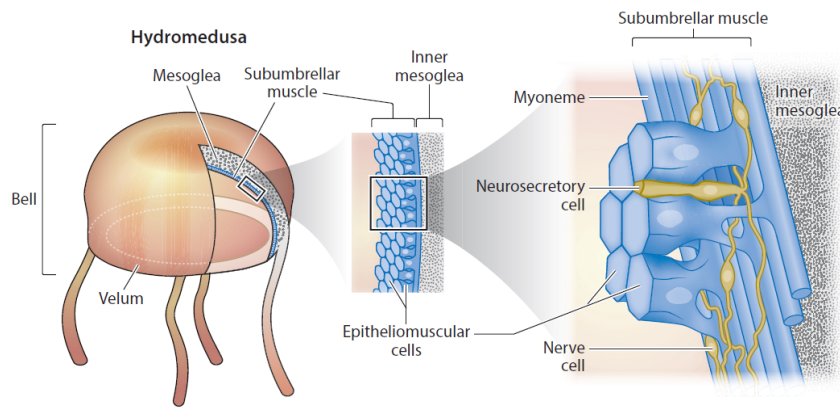


Figure 2: Hydromedusan swimming bell and muscle fiber structures. Figure reproduced from (Costello et al., 2021)

Medusae exhibit coordinated locomotion and interaction with surrounding fluid, while not having a brain, heart or skeleton (Schaadt et al., 2017). Their bodies are radially symmetrical, similar to an umbrella and consist mostly of water and mesoglea. Mesoglea is responsible for structural support and buoyancy, without generating any metabolic losses (Miles & Battista, 2019).

Figure 2 presents the anatomy of a Scyphozoa. The mesoglea, which separates the exumbrellar and subumbrellar surfaces of the bell, is shown. It houses several specialised cell types, including cnidocytes (stinging cells), mucocytes (mucus secretion), sensory cells, and, most importantly for locomotion, epitheliomuscular cells. The gastrodermis, the inner layer of the bell, lines the gastrovascular cavity, where digestion and nutrient absorption occur. It contains nutritive-muscular cells and gland cells, as well as additional cnidocytes (Zachariah & Mitchell, 2006).

The subumbrellar surface at the inner underside of the bell is key to propulsion. It contains circular muscle fibres that can contract to reduce bell volume, leading to a flow of water (a jet), generating thrust. Radial fibres extend from the bell centre to its margin, the edges of the bell. At this location, sense organs are located for sensing light and orienting its swimming, contributing to finer control of bell deformation and directing flow (Gemmell, Troolin, et al., 2015; Zimmerman et al., 2019). Muscle groups consist of epitheliomuscular cells, unique to cnidarians. These groups combine epithelial and muscle functions within a single cell type. Though only one layer thick, they are sufficient to deform the bell due to its low mass. In addition to this low mass, jellyfish must move the mass of surrounding fluid water that is accelerated, representing the added mass (Daniel, 1983; Zachariah & Mitchell, 2006). Cnidarian muscles serve two purposes: (1) contractile functions to assist with locomotion, and (2) intracellular food digestion. The mesoglea contains elastic fibres that help the bell bounce back to its relaxed shape after contraction, enabling passive energy recapture. Neurosecretory cells are involved in transmitting signals for muscle contraction (Costello et al., 2008, 2021; Miles & Battista, 2019; von Montfort et al., 2023). Oral arms, often located near the centre underneath the bell, aid in feeding.

2.1.2 Jellyfish Kinematics

The anatomical constraints of jellyfish, such as minimal muscle mass, in combination with energetically efficient swimming, motivate research into their kinematics. Jellyfish propel themselves through rhythmic bell contractions (Baldwin & Battista, 2021). Regarding the mode of propulsion for locomotion, two groups can be distinguished. A rowing mode of propulsion, used by oblate jellyfish, and a jet propulsion mode, used by prolate jellyfish. Both modes are respectively given in Figure 3 as panels A and B, illustrating a contraction and relaxation phase within the swimming cycle. Jet propulsion is commonly observed in the classes Hydrozoa and Cubozoa (see Figure 1 for reference on these classes), while a rowing mode of propulsion is often found in Scyphozoa. However, there are exemptions as some Hydrozoa use a rowing mode of propulsion or a combination of both modes (Costello et al., 2008). The distinction between prolate and oblate jellyfish is generally applied to define kinematic properties (Colin & Costello, 2002; Costello et al., 2021; Dabiri et al., 2006). Figure 3 shows how prolate jellyfish possess a velum that shapes and accelerates a jet during contraction (B). In contrast, oblate jellyfish lack this structure and instead utilise vortex effects, a rowing mode of propulsion (A). The kinematics of both prolate and oblate jellyfish are further explained in this section.

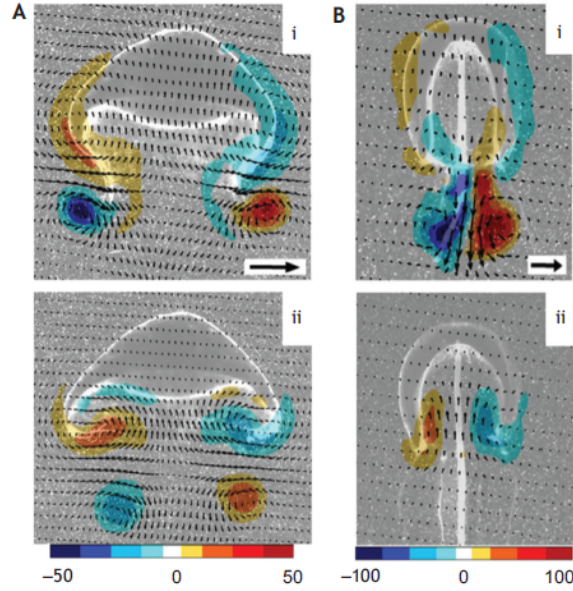


Figure 3: Wake structure vorticity and velocity vectors of (A) Rowing Medusae and (B) Jetting Medusae. Reproduced from (Gemmell, Dabiri, et al., 2021)

Rowing mode of propulsion

Extensive research has been conducted into how oblate jellyfish kinematics efficiently generate thrust (Bajcar et al., 2009; Costello et al., 2021; Dabiri et al., 2005; Gemmell, Du Clos, et al., 2021; Gemmell, Troolin, et al., 2015). Generally, oblate locomotion is described as a pulsation cycle, including a contraction stroke and a relaxation stroke, respectively for power and recovery (Dabiri et al., 2005). The initial state is a relaxed and fully expanded bell, after which the contraction phase starts, and a starting vortex is generated. This contraction phase is short in the context of a full motion cycle and leads to the starting vortex being expelled downstream. During a contraction, the mesoglea stores elastic energy, created by pushing the fluid to the centre and stretching the fibres. This stored energy leads to relaxation of the bell when the muscle tension drops (Pallasdies et al., 2019). During the relaxation phase, which takes

longer, the stopping vortex is formed inside the subumbrellar cavity. This vortex generates a flow into the bell and enhances thrust through passive energy recapture. Interaction of this stopping vortex with the starting vortex induces a wall effect, further explained in Section 2.1.3 (Costello et al., 2021; Dabiri et al., 2005). The ratio of contraction time to full cycle time was defined as the kinematic duty cycle in Baldwin and Battista (2021). Oblate jellyfish exhibit low kinematic duty cycle, with contraction phases comprising only 10-20% of the pulsation cycle. Longer relaxation phases facilitate vortex interaction and passive energy recapture by spacing the starting and stopping vortex appropriately (Baldwin & Battista, 2021; Gemmell, Du Clos, et al., 2021).

The subumbrellar muscles drive contraction, primarily at the bell margin. In the relaxation phase, elastic recoil of the mesoglea restores the bell shape during relaxation (Bajcar et al., 2009; Chapman, 1999). The radial muscles are responsible for directing the flow. The bell margin does not follow the same deformation pattern as the rest of the bell. The differently moving bell margin, often acting semi-passively in response to fluid resistance, contributes to vortex ring geometry and feeding flows (Park et al., 2014; Villanueva et al., 2014).

Jet propulsion

The kinematic cycle of prolate jellyfish begins with a rapid contraction of the subumbrellar cavity, forcibly ejecting water through a narrow velar aperture, created through closing the velum ‘flaps’, as illustrated in Figure 3 B-I. This expulsion generates a high-speed jet and a starting vortex. Following contraction, the mesoglea recoils, and fluid passively refills the cavity during relaxation (Dabiri et al., 2006; Daniel, 1983). In contrast to oblate jellyfish, the duty cycle of prolate jellyfish is around 50%, meaning that the contraction and relaxation phases are similar in length. This induces more frequent, high-velocity pulses, increasing energy costs (Dabiri et al., 2006). The velum plays an important role in reducing these energy costs. It controls aperture diameter during contraction and closes in the contraction phase, increasing pressure in the subumbrellar cavity to accelerate the jet (Daniel, 1983). The aperture expands during relaxation, letting the bell refill and shaping the vortex ring.

2.1.3 Hydrodynamics of Jellyfish

Research has been conducted on both prolate (Baldwin & Battista, 2021; Daniel, 1983; Gemmell, Dabiri, et al., 2021) and oblate (Costello et al., 2021; Dabiri et al., 2005; Park et al., 2014) jellyfish. McHenry and Jed (2003) simulated the hydrodynamics of both paddle and jet mechanisms. Section 2.1.2 described the jellyfish kinematics responsible for certain hydrodynamic effects common in jellyfish locomotion. Most research agrees on three hydrodynamic processes underlying rowing and jet propulsion: suction thrust during bell contraction, passive energy recapture during bell relaxation, and a wall effect boost that occurs following an initial bell contraction (Costello et al., 2021).

Suction thrust is generated through bell contraction. Rearward acceleration of the bell margin, occurring during bell contraction, generates a high-pressure field in the subumbrellar region, while a low-pressure field is formed in the fluid surrounding the bell margin, illustrated in Figure 4. This low-pressure field moves from the bell apex to the margin, while a new high-pressure field arises at the apex (Daniel, 1983). Gemmell, Colin, Costello, and Dabiri (2015) identified that the net force forward results from the low-pressure field and not the high-pressure field in the subumbrellar region, thus implying suction thrust. The reason for this is that the jellyfish utilises the dominant low-pressure field to pull itself forward.

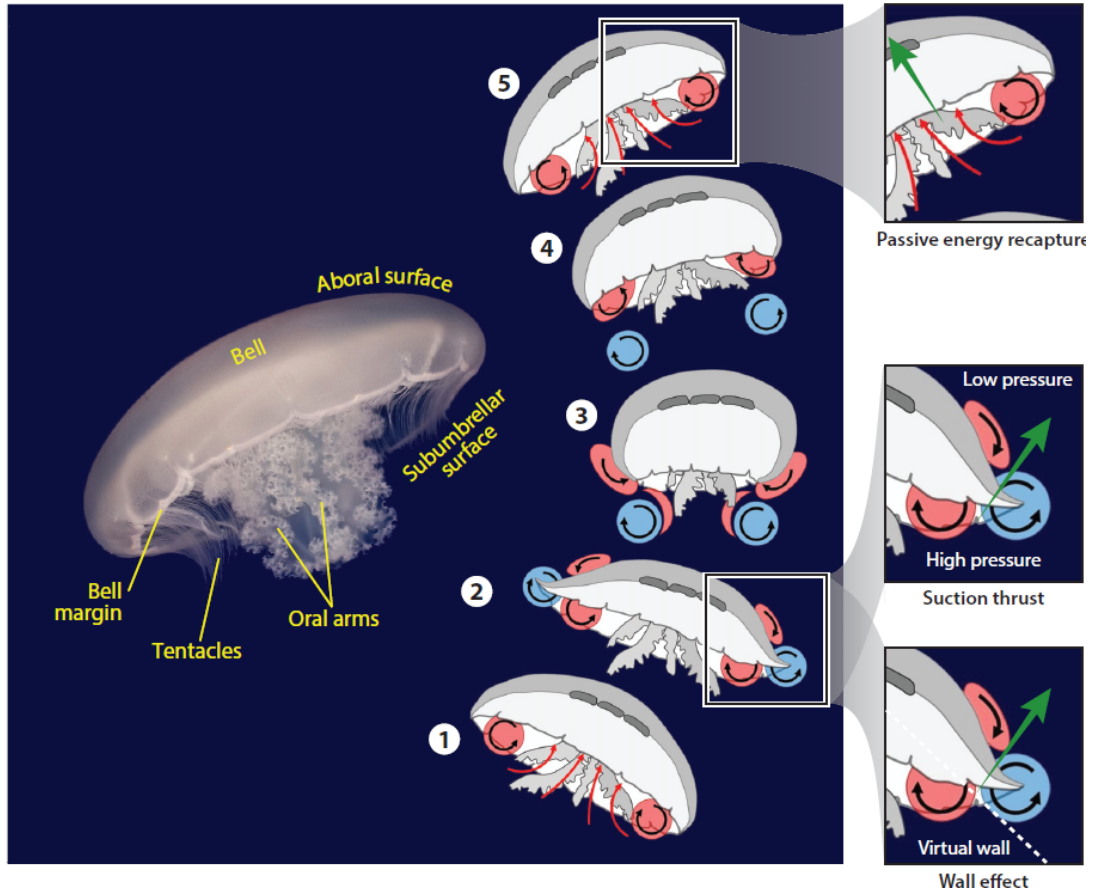


Figure 4: Oblate jellyfish vortex generation. Red areas represent stopping vortices, and blue areas represent starting vortices. Black arrows denote the direction of fluid circulation. Green arrows represent the direction of thrust contribution provided by the different vortex interactions. Figure reproduced from (Costello et al., 2021)

Passive Energy Recapture (PER) occurs during the relaxation cycle, when the subumbrellar cavity fills with fluid. This fluid is part of the stopping vortex generated during bell contraction, which moves into the subumbrellar cavity during expansion. This results in a volume increase in the subumbrellar cavity along with increased fluid velocities, pushing on the subumbrellar surface, creating a thrust force as depicted in Figure 4. Costello et al. (2021) and Gemmell et al. (2018) found that PER depends completely on the duty cycle of bell contraction, as a pause is required to build up this high pressure that can slowly push the bell forward and enable the medusa to ride the vortex-generated pressure gradient. In jetting jellyfish, pulsation cycles are shorter, allowing more rapid animal movement at the cost of being less energy-efficient, as it minimises PER contribution (Gemmell et al., 2018; Hoover et al., 2019). Hoover et al. (2019) added that the elastic properties of the jellyfish bell are impacting PER contribution, being able to increase or decrease it.

The virtual wall effect is another utility of jellyfish to increase thrust and efficiency. This results from opposing vortices that create a ‘virtual wall’. A stopping vortex from the previous pulsation cycle serves in the same way as a virtual vortex during wall effect performance enhancement. The stopping vortex accumulates fluid at the propulsive surface, generating higher pressures at the underside of the bell margin. These high pressures increase the thrust force during the contraction phase (Costello et al., 2021; Gemmell, Du Clos, et al., 2021). This thrust contribution is shown in Figure 4.

A special note should be made on the effect of **added mass**, as this phenomenon can be used to reduce drag. This can be achieved through careful timing of the kinematic cycle to change the volume of a submerged body, recovering added-mass kinetic energy and therefore reducing drag. This phenomenon is further explained in the work of Giorgio-Serchi and Weymouth (2016) and could present itself during jellyfish research.

Rowers produce their forward momentum through the pulsation cycle as shown in Figure 4, implementing the described hydrodynamic effects. They produce thrust during both the contraction phase, through suction thrust and a virtual wall, and the expansion phase, through PER, making them highly efficient (Pallasdies et al., 2019).

Jet propulsion induces similar effects, although its main propulsive force comes from the hydrodynamic phenomenon ‘jet expulsion’, a form of active thrust where pulses of fluid are periodically ejected. Closing the velum reduces the aperture diameter, increasing the pressure in the subumbrellar cavity, inducing a high-velocity flow (jet) through the aperture (Dabiri et al., 2005). A higher pressure arises at the subumbrellar surface, while momentum is sent downstream, and thrust forward is generated. Jet propulsion makes less use of energy recapture due to its ‘50-50’ duty cycle, making jetting jellyfish locomotion less efficient (Costello et al., 2008; Dabiri et al., 2006; Gemmell, Dabiri, et al., 2021). Pulsed jetting is a solution that could increase this efficiency; however, it requires adjusting the duty cycle (Baldwin & Battista, 2021; Costello et al., 2021). However, oblate jellyfish are the more energetically efficient type, although they produce less velocity and induce higher fluid disturbances (Dabiri et al., 2006).

Reynolds number of jellyfish

The Reynolds number of jellyfish depends on the type, whether it is prolate or oblate. Herschlag and Miller (2011) applied Reynolds numbers between 1 and 320 for prolate jellyfish. Colin and Costello (2002) used Reynolds numbers between 1 and 500 and stated that the Reynolds number is closely related to the forward velocity of jellyfish and peaks at the end of bell contraction. Thus, Reynolds numbers can vary during the kinematic duty cycle. Oblate jellyfish have a higher Reynolds number, as they are larger. Baldwin and Battista (2021) found Reynolds numbers between 3000 and 5000 for large oblate jellyfish, while Park et al. (2014) used a Reynolds number of 425 for the oblate jellyfish *Aurelia aurita*. A lower Reynolds number results in a laminar flow instead of a turbulent flow, which simplifies the fluid dynamics. This could make a flow solver more efficient and accurate.

2.2 Validation of the Numerical Jellyfish Model

2.2.1 Overview of modelling approaches

Jellyfish propulsion has been extensively studied across various disciplines using a range of analytical, experimental, and computational approaches. Costello et al. (2021) stated that jellyfish mechanics are not easily modelled, and incorporating detailed biological information is key for successful computational fluid dynamics, instead of relying on assumed kinematic or material traits.

Jellyfish Morphology Modelling

Current jellyfish modelling commonly simplifies bell morphology through applying hemi- or semielliptical profiles. For example, Baldwin and Battista (2021) and Hoover and Miller (2015) adopted a semi-elliptical profile parameterisation as in Equation 1, where (x_c, y_c) is the centre of the ellipse, a is the length of the halfwidth of the bell, and $b + d$ is the height of the bell. Daniel (1983) and Park et al. (2014) added the third dimension through the use of a hemi-elliptical profile parameterisation as in Equation 2. The parameterisations presented here all assume a constant fineness ratio through the contraction cycle, as a and b are not time-dependent. An ellipsoidal parameterisation, although easily applicable in numerical simulation, imposes limitations. A prolate jellyfish may resemble a hemi-elliptical bell in its relaxed phase (Dabiri & Gharib, 2003); however, it is significantly different in shape during the contraction phase. According to Bajcar et al. (2009) and Dabiri and Gharib (2003), an oblate jellyfish differs even more in shape, confirmed by Figure 3. A prolate jellyfish undergoes smaller deformations during actuation and is therefore better modelled through a hemi-ellipsoid than an oblate jellyfish (Villanueva et al., 2014).

$$\frac{(x - x_c)^2}{a^2} + \frac{(y - y_c)^2}{b^2} = 1 \quad \text{for } y \geq y_c - d \quad (1)$$

$$\frac{(x - x_c)^2}{a^2} + \frac{(y - y_c)^2}{b^2} + \frac{(z - z_c)^2}{b^2} = 1 \quad \text{for } z \geq 0, \quad (2)$$

Alben et al. (2013) parametrised the bell shape as a tangent angle function depending on the arc length s of the bell and the time t , as given in Equation 3. This determines the geometry with a single control parameter, β , which governs the kinematics of the bell through adjusting the bell shape. Alben et al. (2013) also designed an analytical model assuming the bell to be a cylinder with time-dependent radius.

$$\theta(s, t) = -1.55(1 - e^{-\frac{s}{\beta(t)}}) \quad (3)$$

Hoover and Miller (2015) and Hoover et al. (2017, 2019) implemented a method where prolate jellyfish are modelled as a hemi-ellipsoid in its initial state, but it deforms under mechanical forces, through stiffness and energy transfer. Bell kinematics can now be quantified through material stiffness, mesh deformation, or oscillation frequency, and not only through a time-dependent fineness ratio, thereby increasing the biological accuracy of the model. In the work of Hoover and Miller (2015) and Hoover et al. (2017, 2019), the bell margin could passively deform under fluid forces; it was modelled as a ring of uniform thickness. Villanueva et al. (2014) modelled jellyfish by gathering points from image tracking. A separate, flexible material was used for the bell margin to induce passive margin motion. To quantify this deformation, it applied the curvature of the exumbrellar surface. However, similar to Hoover and Miller (2015) and Hoover et al. (2017, 2019), this can still only capture passive movements. Bajcar et al. (2009) and Dular et al. (2009) fitted 6th-order polynomials to captured video imaging

of the bell margin deformation, to use as input into computational fluid dynamics (CFD) simulations. This enables implicit flap motion. Feitl et al. (2009) and Miles and Battista (2019) included tentacles through Lagrangian points (representing the structure) tethered together by a virtual spring (a similar approach was adopted by Baldwin and Battista (2021)) and lappets at the bell margin. In all three papers, it was concluded that adding these particles induced vortex and boundary layer interactions, impacting hydrodynamics and propulsion. Wilson and Eldredge (2011) used an approach significantly different from others, modelling the jellyfish as a 2-dimensional articulated system of linked rigid bodies connected by hinges. It consists of seven buoyant, rigid elliptical bodies connected by six torsional hinges. Wilson and Eldredge (2011) states it as not biologically realistic, but loosely inspired by jellyfish mechanics, and it is specifically applied to investigate passive-active mechanical trade-offs in jellyfish swimming.

In general, the fineness ratio, the ratio between bell height and diameter as in Equation 4, is utilised as a quantification of the bell shape (Bajcar et al., 2009; Baldwin & Battista, 2021; Hoover et al., 2019; Park et al., 2014). Dabiri and Gharib (2003) suggested that relying on the fineness ratio alone is insufficient to describe bell kinematics and that incorporating the location of maximum diameter may improve kinematics description. Colin and Costello (2002) added a second morphological variable, the velar aperture area ratio. They assumed a circular cross-section and defined the quantification as the ratio between the velar aperture diameter squared and the bell diameter squared.

$$FR = \frac{H_{bell}}{D_{bell}} \quad (4)$$

A limitation across the literature is the implementation of a non-simplified three-dimensional bell model, with a separately moving bell margin. All research applies simplification in some way, either through shape simplification or reducing dimensionality. Detailed morphological accuracy, in terms of having a flap and tentacles, is nonexistent, although several simplified options have been presented and discussed.

Jellyfish Kinematics Modelling

Numerical jellyfish models commonly imitate jellyfish kinematics from experimental findings gathered through image tracking. Coordinated muscle contraction and mesogleal recoil resulted as key kinematics for the thrust generation process and therefore serve as a foundation in kinematics models. These kinematics are governed by the material properties of the mesoglea and the radial muscle fibres. Megill et al. (2005) stated that the fibres alone can account for the energy required to passively refill the subumbrellar cavity during relaxation, which was confirmed through video tracking in Bajcar et al. (2009). Some numerical models incorporate material properties, such as elasticity and stiffness, as well as deformation dynamics, including contraction and recoil, to replicate the kinematics.

Each kinematic model requires a method that can make the jellyfish move. Sahin et al. (2009) modelled swimming hydromedusae by digitising the outline of the medusan bells with 2D non-uniform rational B-spline curves (NURBS) and applying Fourier-series interpolation in time. By directing a laser sheet through the central axis of the medusae and recording its movements in the laser sheet, the hydromedusa locomotion in 2D was derived. The resulting digitised bell kinematics for free-swimming hydromedusae are given in Figure 5, with the parameter T as the period of one cycle and the small t representing which point of the cycle each line corresponds with. Bajcar et al. (2009) used 6th-order polynomials to copy experimentally observed jellyfish kinematics. Data-driven simulations that replicate experimentally observed kinematics

highlight the problem of insufficient modelling of the bell margin motion (Villanueva et al., 2014). Wilson and Eldredge (2011) found that these flaps significantly enhance thrust production. Therefore, having a standardised kinematic model lacking this flap could underestimate the propulsive properties of jellyfish. To account for this, Wilson and Eldredge (2011) described kinematics as sinusoidal changes in hinge angles, based on biological data, thus actively applying motion to the model. Some hinges could be made passively responsive using torsion springs, enabling passive motions. Hoover and Miller (2015) and Villanueva et al. (2014) also described it as passive motion. Other research models the bell margin motion actively. For example, McHenry and Jed (2003) used a sawtooth function to define kinematic equations for the motion of the margin angle, bell height and bell diameter. The margin angle can then be actively adjusted during contraction cycles.

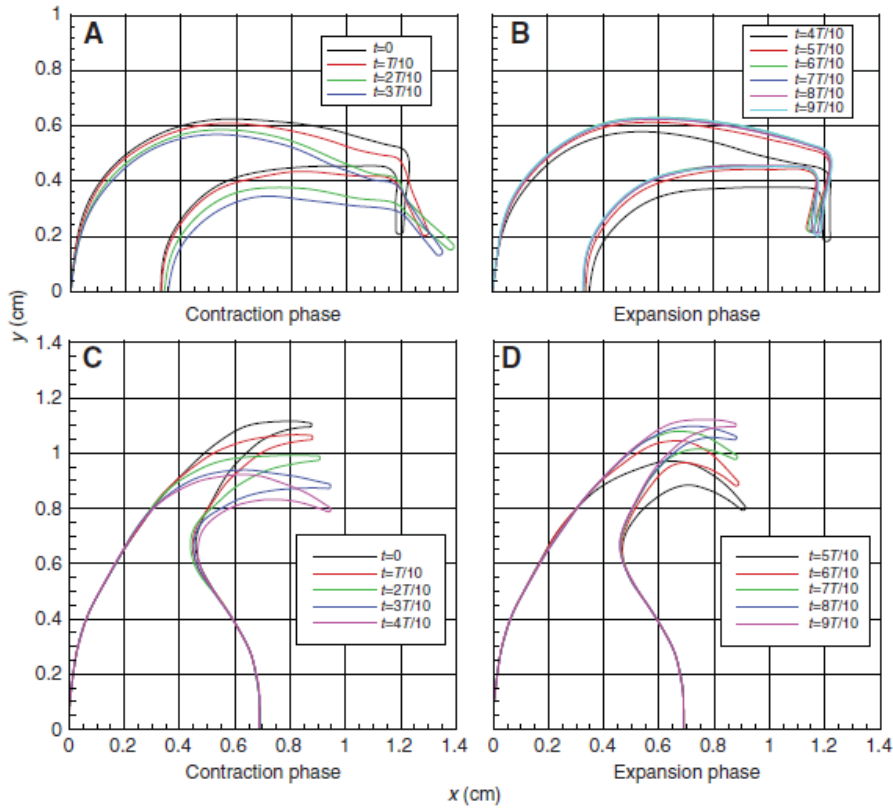


Figure 5: The time variation of the bell morphology for the free-swimming hydromedusae *S. tubulosa* (upper) and *A. victoria* (lower) as reported by Sahin et al. (2009)

Other models also implement the elastic properties of jellyfish. Hoover et al. (2017, 2019) developed a spatially varying model, implementing an elastic modulus gradient in the vertical direction. Hoover and Miller (2015) and Hoover et al. (2017, 2019) initiated the bell's contraction by applying a time-varying tension to the bell margin, while the expansion phase is initiated by removing the applied tension such that the stored elastic energy drives the expansion of the bell. Other reduced-order models, used by Demont and Gosline (1988) and Megill et al. (2005), approached elasticity using spring-mass-damper analogies, creating a system with uniform parameters at specific points. Likewise, Baldwin and Battista (2021) and Miles and Battista (2019) discretised the hemi-ellipsoid and connected the points through virtual springs and beams with individual elastic properties. Baldwin and Battista (2021) applied varying

spring stiffness across points of the discretised hemi-ellipsoid, relative to the preferred geometric configuration (the initial hemi-ellipsoid state), to account for varying elastic properties. It can deform by dynamically changing the spring resting lengths and then initiate contraction. Park et al. (2014) assessed the influence of elastic moduli and found that the bell deformations were affected by the tension modulus but not the bending modulus. The swimming performance was shown to be dependent on the bell elasticity, and an optimal elasticity value was identified that maximised the propulsive efficiency. This illustrates that using a prescribed kinematics model could underestimate the propulsive efficiency of jellyfish. However, applying a passively responsive model requires a fluid-structure interaction (FSI) approach, significantly increasing simulation complexity and time.

While current models increasingly capture the coupling between active muscular contraction, passive elasticity, and fluid dynamics, many are constrained in some way. Some models remain two-dimensional or exclude passive elastic motion, while others assume uniform elasticity and stiffness. A biologically accurate representation of bell margin motion within a fully three-dimensional bell model is not available, although both components have been explored independently. Applying a method where the bell kinematics are digitised into numerical points and connected through NURBS offers an accurate replication of bell motion, including separate bell margin motion. This method does not require an FSI approach, as the kinematics are predefined. The consequence of this is that passive motions are not simulated, so the bell does not move as a result of the flow around it. This complicates capturing the effects of passive energy recapture during simulations.

Dabiri and Gharib (2003) stated that each kinematic assumption is associated with a quantifiable error, which should be accounted for during research. Dabiri et al. (2005) quantitatively evaluated the thrust production mechanisms. They applied the wake vortex ring volume relative to the ejected fluid volume as a quantification of the vortex generation. These volumes were derived from measuring the subumbrellar surface from video analysis. Together with Daniel (1983), the anticipation is made that volume is the most useful index of swimming thrust in field measurements. A recently identified phenomenon involves a reduction in flow velocity caused by vorticity cancellation between starting and stopping vortices, which arises from sub-optimal stroke timing (Hoover et al., 2019). This suggests that some form of stroke timing quantification is required in the kinematics model.

Herschlag and Miller (2011) and Wilson and Eldredge (2011) used a ‘kinematic Reynolds number’ as given in Equation 5, providing a measure of inertial versus viscous effects arising from the motion of the bell. McHenry and Jed (2003) quantified bell kinematics in terms of fineness ratio, margin angle and swimming speed. Dabiri et al. (2006) quantified kinematics through cavity volume and velar diameter as a function of time.

$$Re_k = \frac{\rho U_{body} D}{\mu} \quad (5)$$

Hydrodynamic and Fluid-Structure Interaction Modelling

The swimming kinematics of jellyfish generate thrust through interaction with the surrounding fluid dynamics. These interactions have been modelled in various ways. Daniel (1983) and McHenry and Jed (2003) modelled the sum of thrust, drag, added-mass (acceleration) force, and reaction forces arising from body motion as in Equation 6. McHenry and Jed (2003) calculated these forces through mathematical equations of the bell shape and kinematics. The resulting swimming velocity is computed numerically using a variable-order Adams–Bashforth–Moulton solver. Compared to experimental results, this method results in quite accurate predictions for prolate jellyfish. However, it applies prescribed motions and does not solve for the fluid equations, making it a reduced-order method.

$$T + D + A + F = 0 \quad (6)$$

Other models do include the step of numerically solving simplified Navier-Stokes equations that govern hydrodynamics. Simplifications through assuming an incompressible, viscous flow of a Newtonian fluid are generally used (Bajcar et al., 2009; Hoover et al., 2017; Miles & Battista, 2019; Park et al., 2014). These assumptions are generally acceptable for lower Mach numbers, which is the case for jellyfish (≤ 0.5), as its velocity is significantly lower than the speed of sound through water. The Navier-Stokes equations for an incompressible viscous flow are shown in Equation 7 and 8, respectively representing the conservation of mass and momentum. The mass conservation equation ensures that fluid density remains constant (incompressibility), while the momentum equation describes how fluid motion evolves in response to pressure, viscous, and inertial forces.

$$\nabla \cdot \mathbf{u} = 0 \quad (7)$$

$$\rho_0 \left(\frac{\partial \mathbf{u}}{\partial t} + \mathbf{u} \cdot \nabla \mathbf{u} \right) = -\nabla p + \mu \nabla^2 \mathbf{u} + \mathbf{f} \quad (8)$$

Several numerical solvers are available to compute the Navier-Stokes equations, which will be addressed below. In addition to the fluid model, the structure also requires modelling, depending on the applied approach. In structure modelling, two main approaches are used: kinematic and dynamic. Kinematic models, used in Bajcar et al. (2009), prescribe motion directly and apply it to the fluid without solving structural equations. Thus, only the Navier-Stokes equations need to be solved. Dynamic models, used in Hoover et al. (2017) and Park et al. (2014), solve structural force equations to determine body deformation and response, and couple the results to the fluid equations.

Methods using prescribed motion only require a solver for the Navier-Stokes equations. Alben et al. (2013) utilised a 2D vortex-sheet model to simulate the fluid flow around the jellyfish bell during swimming. This model accurately captures vortex dynamics. They assumed an inviscid flow and used a bound axisymmetric vortex sheet as a boundary condition for the bell surface, representing a discontinuity in fluid velocity. For the analytical assumption-based bell model, they applied a potential-flow model, thus assuming no vorticity. Bajcar et al. (2009) applied a 2D SIMPLE algorithm, described by Baliga and Patankar (1980) as a convection-diffusion problem solver. This method divides the domain of interest into three-node triangular elements. Control volumes are generated between triangle centroids and used in conservation equations. Baliga and Patankar (1980) stated that this method is mainly applicable for 2D problems. Sahin and Mohseni (2009) applied a 2D geometrically conservative arbitrary Lagrangian-Eulerian formulation, obeying the geometry conservation law. In this method, the mesh follows the interface between the fluid and solid boundary and the governing equations

are discretised on an unstructured moving and deforming mesh. Gemmell, Dabiri, et al. (2021) stated that this method enables control of parameters of the numerically simulated organisms, and isolating specific effects becomes possible. Oppositely, Sotiropoulos and Yang (2014) stated it is cumbersome for problems with large deformations. Wilson and Eldredge (2011) used a 2D viscous vortex particle method to solve the fluid flow for its prescribed motions model. According to Eldredge (2007), this method solves the Navier-Stokes equations using a fractional step procedure. Vorticity is carried by particles that interact with one another, exchanging strength to simulate convection and diffusion. Slip along the body is corrected by introducing a vortex sheet and diffusing it into nearby particles. Particles are regularly remeshed to a Cartesian grid to maintain consistency. The Kelvin’s circulation theorem is enforced by recovering circulation lost to the body. The method is validated with insect-inspired flapping and fish-like swimming cases (Eldredge, 2007).

The methods that assume passive bell motion generally apply FSI methods, where the structural equations should be solved in addition to the Navier-Stokes equations. The Immersed Boundary Method (IBM) is a fluid solver framework. An example of the method is described by Sotiropoulos and Yang (2014) and used in Hoover et al. (2017) and Park et al. (2015), which solves the Navier-Stokes equations on a fixed background grid. The method represents structural interactions through independent surface meshes that are accounted for in the governing equations of the grid. This method couples the two by applying forces from the structure to the fluid. The fluid and structure equations can be solved through different solvers within this framework. Hoover et al. (2017) applied an IBM where FSI is described in an Eulerian frame and structural deformations and stresses in a Lagrangian frame. A finite difference formulation is used for fluid modelling and a finite element formulation for structural modelling in 3D, allowing for passive structural deformation. Park et al. (2015) solved the fluid governing equations using the fully implicit decoupling method as described by Kim et al. (2002). Velocity and pressure are decoupled and solved separately, allowing for a more efficient solution to the problem. The solid motion equation is derived by the energy method and solved using an iterative method. Again, the fluid is solved in an Eulerian domain and the structure in a Lagrangian domain. Park et al. (2015) applied the no-slip condition at the boundaries of the computational domain. The domain was scaled with the diameter of the jellyfish. They conduct convergence studies by assessing the fineness ratio and velocities for different domains and grid sizes. A 3D penalty IBM is used in Park et al. (2014). This method separates a massless and a massive part of the immersed boundary and links them through a virtual (stiff) spring with damping. This spring is given a ‘penalty’ force to enforce that the massive and massless parts have the same location and velocity. Park et al. (2014) used a fractional step method to solve the fluid motion equation, which decouples coupled terms and divides time steps into fractional time steps to improve computation. It uses the subdivision finite element method, described in Cirak et al. (2000), to solve the solid motion equation. The equations are coupled by adding a momentum force term.

Baldwin and Battista (2021) and Herschlag and Miller (2011) used the IBM developed by Peskin (2002) in 2D. Implication of the IBM, as described by Peskin (2002), applies an Eulerian frame for the fluid and a Lagrangian frame for the structure. These frames are coupled through implicitly enforcing the dynamic and kinematic boundary conditions on the immersed structure through Equation 9 and 10. Equation 9 enforces the no-slip condition by making the structure velocity, $\frac{\partial \mathbf{X}}{\partial t}(s, t)$, match the local fluid velocity on the structure location. Equation 10 describes the force of the structure on the fluid through the force per unit length $\mathbf{f}(r, t)$, implicitly representing the dynamic boundary condition through a force sum. In both equations, a 2D Dirac

delta function is used to translate quantities between the frameworks, and $\mathbf{s}$ is a state function representing the kinematic state.

$$\frac{\partial \mathbf{X}}{\partial t}(r, t) = \int \mathbf{u}(\mathbf{x}, t) \delta(\mathbf{x} - \mathbf{X}(r, t)) d\mathbf{x} \quad (9)$$

$$\mathbf{F}(\mathbf{x}, t) = \int \mathbf{f}(r, t) \delta(\mathbf{x} - \mathbf{X}(r, t)) dr \quad (10)$$

WaterLily.jl, an incompressible viscous flow solver, uses the Boundary Data Immersion Method. It is implemented as described by Weymouth and Font (2025) and can be applied to 2D and 3D problems. It is an incompressible viscous flow solver that implements the finite volume method. The solver does not apply a structure solver that can account for the passive elastic behaviour of the structure. Instead, it uses prescribed motion by applying signed distance functions and mapping of the solid boundary to a new location every time step. Thus, motions are defined through functions instead of being a result of a structural equation. The fluid velocities at the solid boundary are equalised to the solid velocities, creating a single equation that is valid over the entire space. The benefit of this method is the possibility of 3D simulations. A limitation of this method is that elasticity cannot be modelled as the solid equations are not implemented in the method. It would require prescribed kinematics to work. This simplifies the problem significantly, as no FSI has to be computed, while accuracy may be preserved by digitising actual jellyfish kinematics from experimental work.

A typical quantification of flows is the Reynolds number. It can be used to determine the flow regime of a fluid and represents the balance between inertial and viscous forces. Park et al. (2014) stated that Reynolds numbers of 50 to 500, a state in which both inertial and viscous forces are present, are representative of natural jellyfish swimming. Herschlag and Miller (2011) agreed with this and stated that organisms do not use reciprocal methods of locomotion below a Reynolds number of $\mathcal{O}(1)$. They define the Reynolds number as a function of the average forward velocity, given in Equation 11.

$$Re_m = \frac{\rho U_{avg} D}{\mu} \quad (11)$$

Dabiri et al. (2005), Park et al. (2015), and Sahin et al. (2009) presented a formulation for the vortex formation time as in Equation 12. This indicates the vortex formation process by the ejected fluid. It is the fraction between jet velocity and diameter integrated over time.

$$t^* = \int_0^t \frac{U_e(\tau)}{D_e(\tau)} d\tau \quad (12)$$

Dabiri et al. (2006) applied vortex formation time/number as defined, but expresses it in measured terms. It also presents a governing equation to derive a locomotive force estimation, which is later used for quantification. This equation is given in its most general form in 13, where $\mathbf{I}$ is the identity matrix, C_{AM} is the added mass coefficient, U_V is the velocity of the vortex wake, and V_v is the volume which can be calculated from the ejected fluid.

$$\mathbf{F}_L = -\rho \frac{d}{dt} [(\mathbf{I} + \mathbf{C}_{AM}) \mathbf{U}_V(t) V_V(t)] \quad (13)$$

Modelling Assumptions and Limitations

To create a numerical model for which the structure and fluid dynamics can be solved, several assumptions and simplifications have to be made, both in kinematic and hydrodynamic modelling. This limits the accuracy and reliability of the results. Dabiri and Gharib (2003) assessed the influence of kinematic approximations on the accuracy of swimming dynamics in jellyfish modelling. It states that the approximation of the aperture area is significantly less influential than the approximation of the bell volume. It also states that the hemi-ellipsoid assumption is only valid in the relaxed state. Dabiri et al. (2006) found that a constant velar aperture would result in significant backwards motion of the animal during the refilling phase, and is therefore not a valid assumption.

A key limitation in studies such as Bajcar et al. (2009) is the implementation of prescribed motion, which prevents the model from capturing structural feedback resulting from fluid forces. Passive elements such as tentacles and oral arms are omitted in most models to avoid their negative influence on vortex generation (Feitl et al., 2009; Miles & Battista, 2019). Two-dimensionality is a limitation of models such as Baldwin and Battista (2021), Miles and Battista (2019), and Wilson and Eldredge (2011). Herschlag and Miller (2011) stated that vortex dynamics differ between a 2D model and the organism (for oblate jellyfish).

Bajcar et al. (2009), Hoover et al. (2017), Miles and Battista (2019), and Park et al. (2014) all assumed an incompressible, viscous flow of a Newtonian fluid. Bajcar et al. (2009) also assumed a laminar flow so that no averaging or filtering of the Navier-Stokes equations is required. Herschlag and Miller (2011) stated that the IB method by Peskin (2002) is limited in resolving the shedding of vortex structures for $Re > 1000$. Alternative methods that better handle sharp boundaries at higher Re would be valuable for exploring larger-scale dynamics. Sahin et al. (2009) assumed swirl-free axisymmetric Navier-Stokes equations. Dabiri et al. (2006) do not include resistive fluid dynamic forces in their model, as the required measurements of the oral cavity and velar kinematics could not be obtained from a view that contained the full exumbrellar surface. The outer surface profile is necessary to estimate both viscous drag and the acceleration reaction (added mass) of the animal body. An assumption that has been experimentally supported is that vortex ring formation does not affect the refilling phase in hydromedusae; therefore, the refilling flow may be treated as a uniform inflow Dabiri et al. (2006) and Daniel (1983).

2.2.2 Model Validation

Multiple studies validate their numerical jellyfish model using experimental data of shape, kinematics and hydrodynamics. This data is often extracted from video analysis, direct mechanical testing or flow visualisation techniques. The data serves as benchmarks for assessing the accuracy of model geometry, input motion, and fluid response.

Bajcar et al. (2009) applied image processing techniques on video recordings to capture the bell contour and movements of scyphomedusae. The authors analysed the velocities of selected points along the bell contour and presented these results. Similarly, Dabiri and Gharib (2003) developed a pixel-based search algorithm to reconstruct scyphomedusan shape and motion from video frames. Gemmell et al. (2013) used video recordings of scyphomedusae to determine swimming kinematics. McHenry and Jed (2003) measured bell diameter, height, margin angle, and swimming speed from recordings of live jellyfish. Park et al. (2014) used these data to validate a 3D numerical model. In addition, Park et al. (2014) selected material parameters such as tension and bending moduli, based on the mechanical measurements reported by Demont and Gosline (1988) and Megill et al. (2005). Demont and Gosline (1988) presented static mechanical properties of the bell structure of hydromedusae through non-destructive testing of the mesoglea. Megill et al. (2005) applied experimental sample testing to gather mesoglean fibre properties of a hydromedusan jellyfish.

Hydrodynamic validation involves comparing vortex patterns and swimming forces/velocities. Gemmell, Troolin, et al. (2015) used 3D particle image velocimetry to visualise 'footprints' of the organism, enabling wake dynamics to be captured in high spatio-temporal resolution. It states that 2D particle image velocimetry has the problem that thrust and drag cannot be separated, while the body experiences both thrust and drag. Separation would require a 3D visualisation. So one can see flow accelerations around the bell, but it cannot be said how much is thrust and how much is drag. Flow visualisation data from Dabiri et al. (2005) and Dabiri et al. (2006) provided detailed field and lab measurements of scypho- and hydromedusae. These include dye visualisations of vortex ring structures, bell shape evolution, and flow separation. Wilson and Eldredge (2011) used this experimental data to recreate kinematics to simulate and compare vortex formation and thrust in a bioinspired model. Alben et al. (2013) used the experimental data of Dabiri et al. (2006) (bell diameter) to compare the simulation results of its vortex-sheet model. Additionally, they compared the results with other numerical simulation results, using the immersed boundary method. Park et al. (2014) validated the bell fineness ratio and velocity with experimental data presented by Colin and Costello (2002). This validation is given in Figure 6. Colin and Costello (2002) used video recordings of several hydromedusae to determine morphological and kinematic parameters. The study also presents hydrodynamic data in the form of particle velocities relative to the bell.

Hoover et al. (2017) validated their numerical model against biological measurements of a scyphomedusa and a hydromedusa, using data from Demont and Gosline (1988) and Gemmell et al. (2013). The latter provided quantitative data on swimming speed, distance travelled, circulation, bell diameter, and axial forces. Likewise, Baldwin and Battista (2021) compiled a dataset of jellyfish species, reporting parameters such as bell diameter, fineness ratio, frequency, Reynolds number, and duty cycle, alongside numerical simulation outputs including vorticity maps and swimming velocities.

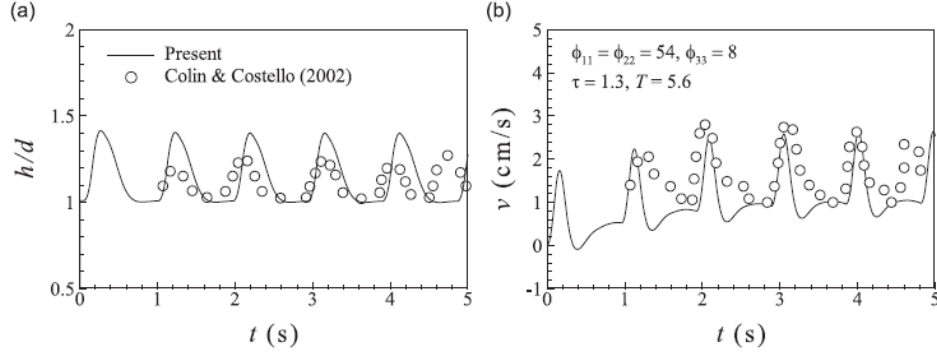


Figure 6: Time histories of (a) the fineness ratio and (b) velocity, as determined by simulation in Park et al. (2015) or measured experimentally and reported by Colin and Costello (2002)

Other studies, such as Gemmell, Dabiri, et al. (2021) and Gemmell et al. (2018), provided vorticity and velocity field data across jellyfish types, useful for validating hydrodynamic results. Gemmell et al. (2013) quantified fluid motions through 2D digital particle image velocimetry and conducted a sensitivity study of mesh and time step refinement. Sahin and Mohseni (2009) validated their numerical model for the steady and oscillatory flow around a sphere in a cylindrical tube, as experimental data of the specific jellyfish type considered was missing. They validated their wake structures with the dye visualisation results of Dabiri et al. (2005, 2006), showing consistency. Swimming kinematics were validated with data from Colin and Costello (1996) and Colin and Costello (2002). Finally, Herschlag and Miller (2011) cautioned that while flow profiles and swimming speeds in 2D numerical models agree well with experimental data for prolate bells, such models fail to capture the dynamics of oblate geometries. This highlights the importance of selecting appropriate validation targets that match the dimensionality and morphology of the numerical model.

Generally, research is validated through experimental results or field data of specific jellyfish types. Data is available for various jellyfish types, both prolate and oblate. Papers often present validation visually with a graph of the assessed parameter. Validation data for shape, kinematics and hydrodynamics is available, making biological-accurate modelling a viable option. To validate the kinematic model in this research, the data given in Sahin et al. (2009) can be used. This data is presented in Figure 5, showing the contraction and expansion phase kinematics for both a prolate and oblate jellyfish type. In this report, results are shown regarding the vorticity fields, velocity components and force coefficients, which can be used to validate the results from the numerical jellyfish model. However, the aim is to validate the jellyfish model with experimental data, presented in Figure 6 and given in Colin and Costello (2002).

2.3 Control Problem of the Numerical Jellyfish Model

In the preceding sections, the control problem of the jellyfish bell margin was discussed. It is questioned whether jellyfish control the bell margin actively or passively. It is also questioned which control mode would result in the most efficient propulsion (Wilson & Eldredge, 2011). Passive control is computationally more intensive, as the problem requires FSI solvers. With active control, solid motions can be directly assigned to the fluid motion equations. Considering that bell margin motion may involve (semi-)passive behaviour, it may be hard to employ active control in a biologically accurate way. Exploring the field of flow control and its currently used applications can be beneficial for designing a representative numerical model that controls the flow.

2.3.1 Flow Control Options

Passive Flow Control, as described by Joshi and Gujarathi (2016), involves no external energy input, and active flow control requires energy and involves a control system. An example of a passive method is the Gurney FLAP, which is perpendicular to the fluid flow direction, resulting in the formation of two counter-rotating vortices that help delay flow separation, therefore reducing drag. Passive techniques modify the base flow, making it more stable, suppressing and delaying disturbance/instability growth mechanisms. Dobrinsky and Collis (2000) stated that typically, the more stable a flow, the more receptive it is to disturbances. Active flow can target small disturbances directly. Collis et al. (2004) added that the ability of a passive device to control unsteady motions is limited, due to instabilities and turbulence.

All passive controls are effectively open-loop. In open-loop control, either steady (passive) or unsteady (active) actuator parameters are set at the design stage and remain fixed. Closed-loop control requires a controller that can withstand unexpected environmental disturbances or uncertainties in the models used to design the controller. The disturbances in flow systems, accompanied by the approximations made to predict fluid flows, make accurate control design difficult (Collis et al., 2004).

Active Flow Control is described by Collis et al. (2004) as a method of altering a natural flow state or development path into a more desired state (or path). Joshi and Gujarathi (2016) classified AFC into two categories: open- and closed-loop. Open-loop methods involve the introduction of steady or unsteady energy inputs without considering the state of the flow field. Closed-loop methods continuously adjust the input to actuators based on sensors. Open-loop is generally used in computational and experimental studies, while closed-loop is less commonly used, due to its complexity. Open-loop systems use a pre-defined actuator input that does not adapt to the flow state. Closed-loop systems use sensors and a control algorithm to adapt actuator input in real time (Collis et al., 2004).

Every active flow control (AFC) problem starts with a linear or non-linear dynamical system. Generally, linear systems are an idealisation of non-linear real-world systems, especially in terms of fluid systems. In practice, linear control is often successfully applied to non-linear problems (Collis et al., 2004). Collis et al. (2004) stated that optimal control can be used to construct local-optimal open-loop control distributions for the full non-linear equations of motion. Optimal control, however, requires a full mathematical model of the system. To acquire such a model, ‘simplifications’ are required to make the state equations solvable. Reduced-order models that capture the essential physics of the flow while reducing the expense associated with solving the model offer a solution.

2.3.2 Applications of Flow Control

Joshi and Gujarathi (2016) provided applications of active flow control, including optimising flow field distribution, increasing lift, and reducing drag. Considering the previously discussed jellyfish models, an example of active flow control is the application of a prescribed bell motion (Bajcar et al., 2009). The kinematics are prescribed through 6th-order polynomials with coefficients updated at each simulation time step. Similarly, Alben et al. (2013) used prescribed parametric equations in the form of power-law functions of time, and Herschlag and Miller (2011) prescribed contraction and expansion kinematics by describing the force that the boundary applies to the fluid.

Across the jellyfish models, claims regarding non-linear vortex interactions (vortex (rings) generation and shedding) (Gemmell, Dabiri, et al., 2021), time-dependent parameterisations and contraction cycles (Alben et al., 2013; Hoover et al., 2019; Park et al., 2014) characterise an unsteady non-linear system. The applicable fluid state equations to the assessed jellyfish problems are the unsteady Navier-Stokes equations (Park et al., 2014). These are non-linear partial differential equations. In several discussed papers, physical reduced-order modelling is applied, accomplished by neglecting certain physical processes that can potentially occur; these are the assumptions described in subsection 2.2.2. Collis et al. (2004) presented another example of ROM, data fitting, which is also used in previously discussed papers. Jellyfish shape and kinematics were fitted to data available from experimental research (Alben et al., 2013; Bajcar et al., 2009; Herschlag & Miller, 2011; Sahin et al., 2009). Collis et al. (2001) presented an example of numerically solving optimal control problems governed by the compressible Navier-Stokes equations. In this study, the control mechanism is suction and blowing in the wall normal direction, and it is applied as a function in the boundary condition equations.

Deep reinforcement learning (DRL) can offer a solution to the flow control problem and as a shape optimisation option. DRL has previously been used for high-level control of large systems, such as the swimming of a school of fish (Wang et al., 2023), and for detailed active flow control at the small scale of flow instabilities and separation (Rabault et al., 2020). It is stated that reinforcement learning is significantly more applicable to closed-loop flow control systems. Mecanna et al. (2025) researched the performance of various DRL algorithms in fluid dynamics problems and showed that Proximal Policy Optimisation has established dominance over other algorithms across a wide range of benchmarks. Rabault et al. (2019) studied a circular cylinder with $Re = 100$, in the laminar flow regime, which is later compared to a case with $Re = 1000$, in the low-turbulence flow regime in Ren et al. (2021). For the first case, successful implementation of DRL was already achieved, whereas for the last case, it was newly achieved in Ren et al. (2021). As this is the order of magnitude of the Reynolds number characteristic of the jellyfish case, the application of DRL may be possible for shape optimisation or active flow control. The application possibilities of reinforcement learning are assessed for the jellyfish control problem, but implications will not be an objective in this research.

2.3.3 Definition of efficiency for a swimming jellyfish

Quantifying the swimming efficiency of jellyfish is difficult as a result of unsteady behaviour, entanglement of thrust and drag, the large range of bell shapes and sizes (Gemmell, Dabiri, et al., 2021). Entanglement of thrust and drag results from the same propulsive surfaces contributing to either thrust or drag, depending on the phase of the swimming stroke. As a result, various definitions of performance metrics have been proposed across the literature. Generally, efficiency is described through the swimming velocity and cost of transport (COT).

$$\text{COT} = \frac{\text{Energy}}{\text{Mass} \times \text{Velocity}_{avg}} \quad (14)$$

Gemmell et al. (2013) used COT as the performance metric for efficiency, defined as a function of energy, mass and average velocity in Equation 14. Gemmell, Dabiri, et al. (2021) stated that this COT reflects energy consumption for its locomotion. However, because this measure has dimensional units, it may not scale isometrically and thus is of more limited utility in comparing across ontogeny or in animals with significantly different sizes or physiology. The observation that one animal group has a lower COT than another might be attributable to differences in their hydrodynamics, physiology, relative size or all three.

In Baldwin and Battista (2021), the performance metrics used are dimensionless. It uses a power-based COT and a swimming speed represented by the inverse Strouhal number, respectively defined as in Equation 15 and 16. The Strouhal number consists of V_{dim} , f and D , respectively the dimensional swimming speed, bell actuation frequency and the maximum diameter during an actuation cycle. In the COT, F_j and U_{rj} were the bell's applied contraction force and contraction velocity at the j^{th} time step, and V_{dim} was the temporally-averaged dimensional forward swimming speed during the N time points considered. The term $\frac{1}{\rho f^2 D^3}$ non-dimensionalises the data, providing a COT metric normalised by the bell's driving frequency and size, and the fluid density. According to Gemmell, Dabiri, et al. (2021), a dimensionless COT is a suitable metric for interspecific comparisons of swimming efficiency. Hoover et al. (2019) also used COT and the Strouhal number to quantify the swimming efficiency of jellyfish.

$$\text{COT} = \frac{1}{N} \cdot \frac{1}{\rho f^2 D^3} \cdot \frac{1}{V_{dim}} \sum_{j=1}^N |F_j| |U_j| \quad (15)$$

$$St = \frac{fD}{V_{dim}} \quad (16)$$

Park et al. (2014) stated that the Froude efficiency is widely adopted as an efficiency metric. It is defined by Wilson and Eldredge (2011) as in Equation 17, the ratio between thrust multiplied by velocity and the input power. However, Gemmell et al. (2013) and Sahin et al. (2009) stated that this quantification overlooks passive energy recapture phenomena and ignores viscous effects, as the relaxation phase is not included in this protocol.

$$\eta_{froude} = \frac{TU}{P_{in}} \quad (17)$$

Sahin et al. (2009) stated a power coefficient as a quantification for swimming performance, as given in Equation 18. This is a non-dimensionalisation of the power required by the hydromedusae assessed in this paper. It is non-dimensionalised by dividing it by the fluid density, average jellyfish velocity and the maximum bell diameter.

$$C_p = \frac{2P}{\rho U^3 \pi (\frac{D}{2})^2} \quad (18)$$

3 Methodology

A clear methodology is essential to answer the stated research question. This includes a description of the steps that need to be taken and the software that will be used. The applied research strategy combines design, validation and experimental (simulation-based) research.

1. The first step in the research process will be to model the bell morphology and kinematics accurately using validation data from experimental research. The aim is to start modelling the bell shape with 2D NURBS (non-uniform rational B-splines). Prescribed motion is applied to make the bell margin move differently from the bell shape. The kinematics should be reconstructed from and validated with the experimental data from Sahin et al. (2009) presented in Figure 5.
2. The validated 2D model should then be extended to 3D. 3D NURBS could be used to acquire such a model. Validation of this model could be difficult, as 3D experimental data of the kinematics is not available. However, if the 2D model is fully validated, the kinematics of the 3D model can be validated from the 2D model kinematics.
3. Using WaterLily.jl, further explained in subsection 3.1, as a CFD solver, hydrodynamic analysis of the swimming 3D model will be conducted. The results from this analysis can then be validated with the data presented in Colin and Costello (2002), similar to Figure 6. This paper presents experimental data in the form of velocity, acceleration and position during the kinematic cycle, which can be used to validate results from the 3D model.
4. Through evaluation of the hydrodynamics that result from the CFD simulations in WaterLily, performance metrics, such as a non-dimensional COT and forward velocity, can be determined. The actively controlled bell margin kinematics can be varied, and a note on the influence of its kinematics on the propulsive efficiency can be made.

3.1 WaterLily as CFD solver

WaterLily.jl is an incompressible viscous flow solver in which the Immersed Boundary method is used to enforce the effect of solid boundaries on flow past complex geometries with arbitrary motions. The applicability to complex geometries makes this solver particularly beneficial for the jellyfish problem, as the bell shape is complex to describe. WaterLily.jl is an open-source code written in the programming language Julia and supports CPU and GPU execution. It implements the governing equations using a finite-volume approach on a uniform Cartesian grid with staggered velocity-pressure variable placement. It is typically restricted to 2D and 3D flows. In Weymouth and Font (2025), various example cases are given, including a parameterised jellyfish.

3.2 Risks and Limitations

Kinematics hard to model

If the kinematics become too difficult to model through the use of NURBS, it would mean that a different modelling approach is required. This means that time is lost in the process of setting up the NURBS model and then switching towards a different modelling method. This risk could be mitigated by applying a method that is also used in the available literature.

Active control not biologically accurate

In case active control of the bell margin does not give an accurate description of the actual

bell motions, this could cause misjudgment of the resulting hydrodynamics and efficiency. This would mean that a switch has to be made towards a passively moving bell margin, which could be time-consuming as it becomes a significantly more complicated model. A mitigation of this risk could also involve applying methods that are available in the literature, such as the method presented in Wilson and Eldredge (2011). However, the model would then become more simplified and therefore less biologically accurate.

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